

Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification | PES University, Dept. of CSE | Software Engineering

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We chose Layered Architecture for the Patient Health Record Consent Management System.

Architectural Choice. Layered architecture with three layers and seven components: Presentation (Patient & Doctor Portal); Business (Authentication Service, Consent Manager, Emergency Access Service, Notification Service, Audit Trail Service); and Data (Encrypted Data Store). The handout's Order Manager and Payment Service became our Consent Manager, which coordinates grant, revoke and access requests, and our Authentication Service, which it calls to verify the clinic doctor. The components are joined by 11 interfaces (API calls between the Portal and Business Layer, database queries into the Data Layer), and requests flow from the Portal down to the data.

Reason 1: one enforcement path for consent and audit. Every grant, revoke and emergency override passes through the Consent Manager, which records it through the Audit Trail Service (IAuditLog). The Emergency Access Service creates its break-glass grant through the Consent Manager (IGrantProcessing), so even an override cannot skip the audit entry required by FR-006 and NFR-001. Expiry (FR-001) and revocation (FR-002) are likewise enforced in a single place.

Reason 2: components map directly onto our Lab 2 backlog. SCRUM-11 (Consent Granting & Access Requests) maps to the Consent Manager and Authentication Service (SCRUM-17 doctor verification); SCRUM-12 (Consent Revocation & Expiry Notifications) to the Consent Manager and Notification Service; SCRUM-13 (Audit Trail & Transparency) to the Audit Trail Service; SCRUM-14 (Emergency Access Override) to the Emergency Access Service and Notification Service (SCRUM-27); SCRUM-15 (Security & Authentication) to the Authentication Service and Encrypted Data Store. All 5 Epics and all 14 stories (SCRUM-16 to SCRUM-29, 68 story points) therefore have a clear home, and a system of this size does not need more machinery than three layers.

Security Advantage. The Portal can reach health data only through the Business Layer, so the Authentication Service (UC-01, SCRUM-28) sits in front of every request and no client ever touches the store directly. All records, grants and audit entries live in one Encrypted Data Store, so AES-256 at rest and TLS 1.2 or higher in transit (NFR-002) are enforced at one boundary, and audit entries are written only through the Audit Trail Service (IAuditData), which keeps the trail append-only (NFR-001).

Performance Benefit. Grants, revocations and emergency flags sit in a single Data Layer, so a revoke is visible to the very next access check with no cross-service synchronisation. This supports the 5-second revocation target (FR-002) and the 30-second emergency grant (FR-006), and audit queries read from one store, which helps the 3-second filter target (FR-005).

Why not the other styles. Microservices: separate deployable services would add network latency, operational complexity and data-consistency risk between the grant store and the audit trail; a stale grant after a revoke could fail FR-002, and the system is too small to justify that cost. **Client-Server:** it only separates clients from one central server; it does not separate consent logic, audit and storage inside that server, which our append-only audit and encryption requirements need, and a single server block is a bottleneck for emergency access. **Trade-off we accept:** with Layered, individual components cannot be scaled independently, which is acceptable for this scope.